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Ex.No. 2: Implementing TDMA Slot Allocation in GSM.

Objective 1:

To calculate the Time Slot Duration in a GSM TDMA frame by dividing the total frametime into equal time slots. This helps in understanding the time structure of GSM communication.

Program (Matlab Code)

```
clc; clear; close all;
FrameTime = 4.615; Slots = 8;
SlotDuration = FrameTime/Slots;
t = 0:SlotDuration:FrameTime;
figure
subplot(2,1,1)
stem(t,ones(size(t)),'filled')
title('TDMA Time Slot Structure')
xlabel('Time (ms)')
ylabel('Slot Level')
grid on
subplot(2,1,2)
plot(t,ones(size(t)),'-o','LineWidth',1.5)
title('Continuous View of Slot Allocation')
```

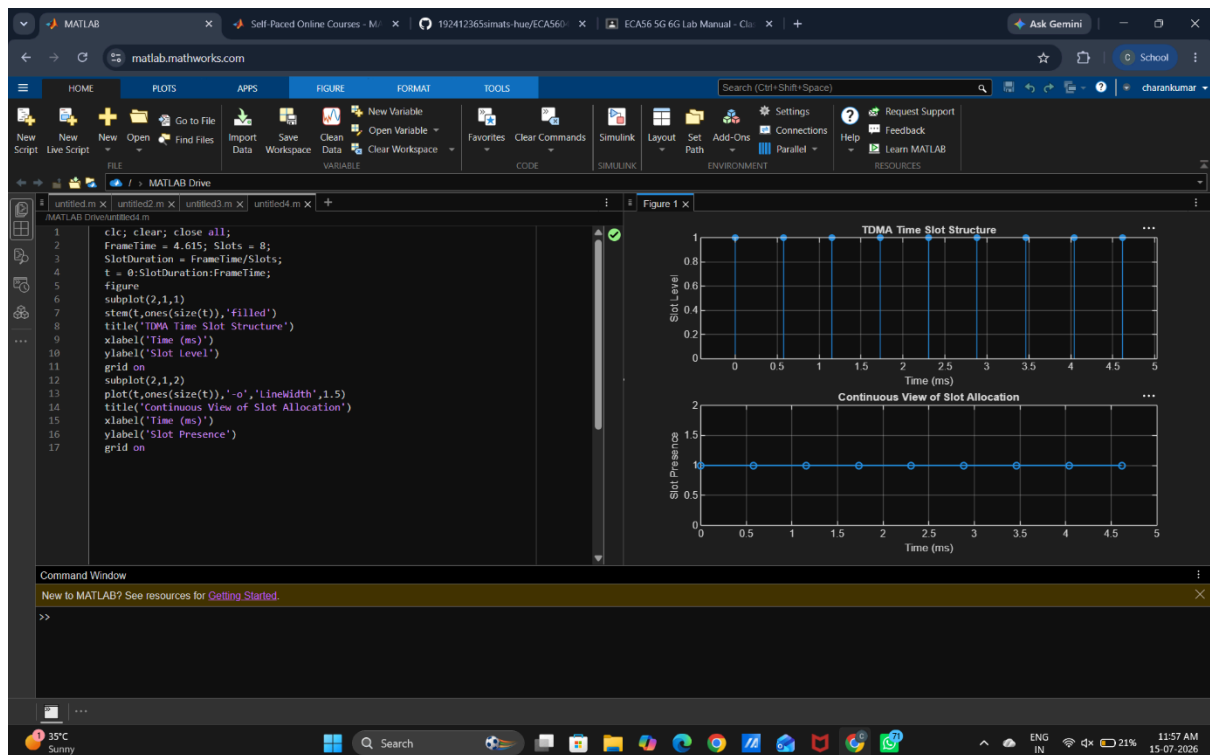


xlabel('Time (ms)')

ylabel('Slot Presence')

grid on

Output:



Objective 2:

To determine the Frame Utilization (%) based on the number of allocated time slots compared to total available slots. This evaluates the efficiency of TDMA resource usage.

Matlab Code

```
clc; clear; close all;
```

```
TotalSlots = 8;
```

```
AllocatedSlots = 6;
```

```
FreeSlots = TotalSlots - AllocatedSlots;
```

```

Utilization = (AllocatedSlots/TotalSlots)*100;

disp('Frame Utilization (%)')

disp(Utilization)

figure

% Graph 1: Simple Line Plot (SAFE - no bar function)

subplot(2,1,1)

plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)

title('GSM Frame Utilization (Allocated vs Free)')

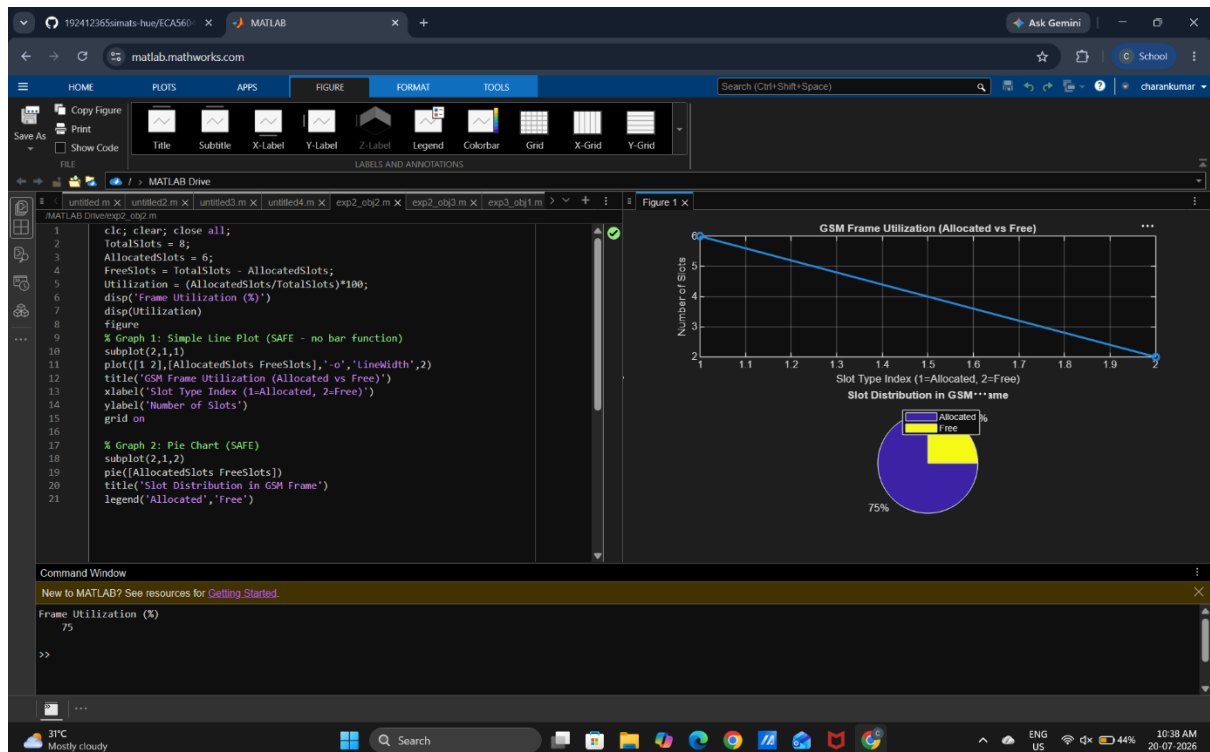
xlabel('Slot Type Index (1=Allocated, 2=Free)')

ylabel('Number of Slots')

grid on

```

Output



Objective 3:

To compute the Number of Allocated Time Slots for multiple users in the GSM system. This ensures proper scheduling and interference-free communication.

Matlab Code

```
clc; clear; close all;

Users = 1:8; % Total users

TotalSlots = 5; % Available slots

Alloc = zeros(1,8); % Initialize allocation

Alloc(1:TotalSlots) = 1; % Assign slots to first users

Blocked = 1 - Alloc; % Remaining users blocked

disp('User Wise Slot Allocation')

disp(table(Users',Alloc',VariableNames',{'User','Status(1=Allocated,0=Blocked)'}))

figure

% Graph 1: Allocation per user

subplot(2,1,1)

stem(Users,Alloc,'filled','LineWidth',2)

title('TDMA Slot Allocation per User')

xlabel('User Index')

ylabel('Slot Status (1=Allocated, 0=Blocked)')

ylim([-0.2 1.2])

grid on

% Graph 2: System view (safe alternative to bar)

subplot(2,1,2)

plot(Users,Alloc,'-o','LineWidth',2)

hold on

plot(Users,Blocked,'-s','LineWidth',2)

title('TDMA Slot Utilization Analysis')
```



```

xlabel('User Index')
ylabel('Status')
legend('Allocated','Blocked')
grid on

```

Output

